Publications

Total publications: 52 / refereed: 40 / first-author: 6 / selected significant contributions: 7 / collaboration papers: 39 / total citations: 8,068 / h-index: 20 (Last updated: 2026-10-31)

First-author publications (citations: 170)

6 **Anand, Abhijeet**; & Muzahid, Sowgat, 2026, *The First Blind Survey of Intervening NaI Absorbers: Incidence and Cosmic Evolution with DESI*, ArXiv, arXiv:2609.26794 (arXiv:2609.26794) **[Impact Factor: N/A; Scopus Quartile: N/A]**

My role: Led the first large blind survey of intervening neutral sodium (Na I) absorbers and measured their low-redshift incidence and cosmic evolution.

5 **Anand, Abhijeet**; Aguilar, J.; Ahlen, S.; Bianchi, D.; Brodzeller, A.; *et al.*, 2025, *The Cosmic Evolution of C IV Absorbers at 1.4 < z < 4.5: Insights from 100,000 Systems in DESI Quasars*, The Astrophysical Journal, **990**, 151 (arXiv:2504.20299) [9 citations] **[Impact Factor: 5.7; Scopus Quartile: Q1]**

My role: Constructed the largest triply-ionized carbon absorber catalogue to date and measured the evolution of its incidence and cosmic mass density over $1.4 < z < 4.5$.

4 **Anand, Abhijeet**; Guy, Julien; Bailey, Stephen; Moustakas, John; Aguilar, J.; *et al.*, 2024, *Archetype-based Redshift Estimation for the Dark Energy Spectroscopic Instrument Survey*, The Astronomical Journal, **168**, 124 (arXiv:2405.19288) [55 citations] **[Impact Factor: 5.8; Scopus Quartile: Q1]**

My role: Led the development and implementation of a novel archetype-based galaxy redshift-fitting algorithm for DESI, significantly reducing catastrophic redshift failures and improving redshift accuracy and reliability.

3 **Anand, Abhijeet**; Kauffmann, Guinevere; & Nelson, Dylan, 2022, *Cool circumgalactic gas in galaxy clusters: connecting the DESI legacy imaging survey and SDSS DR16 Mg II absorbers*, Monthly Notices of the Royal Astronomical Society, **513**, 3210 (arXiv:2201.07811) [33 citations] **[Impact Factor: 5.2; Scopus Quartile: Q1]**

My role: Mapped cool metal-enriched gas around galaxy group and cluster environments and showed that clusters host a substantial reservoir of cool gas in the intracluster medium.

2 **Anand, Abhijeet**; Nelson, Dylan; & Kauffmann, Guinevere, 2021, *Characterizing the abundance, properties, and kinematics of the cool circumgalactic medium of galaxies in absorption with SDSS DR16*, Monthly Notices of the Royal Astronomical Society, **504**, 65 (arXiv:2103.15842) [70 citations] **[Impact Factor: 5.2; Scopus Quartile: Q1]**

My role: Constructed the largest SDSS Mg II absorber sample to characterize the spatial distribution and kinematics of cool circumgalactic gas in and around star-forming and passive galaxies.

1 **Anand, Abhijeet**; Roy, Nirupam; & Gopal-Krishna, 2019, *Search for H I emission from superdisk candidates associated with radio galaxies*, Research in Astronomy and Astrophysics, **19**, 083 (arXiv:1812.06875) [3 citations] **[Impact Factor: 2.6; Scopus Quartile: Q2]**

My role: Analysed VLA observations and led the interpretation and writing of the H I 21 cm study conducted as part of my MS thesis.

Selected significant contributions (citations: 107)

7 Napolitano, Lucas; Myers, Adam D.; Tedeschi, Adam; **Anand, Abhijeet**; Herrera-Alcantar, Hiram K.; *et al.*, 2026, *The Composite Spectrum of QSO Absorption Line Systems in DESI DR2*, The Astronomical Journal, **172**, 250 (arXiv:2512.02992) **[Impact Factor: 5.8; Scopus Quartile: Q1]**

My role: Contributed to absorber selection for the stacking analysis and served as an internal reviewer during the collaboration-wide review.

6 Lan, Ting-Wen; Prochaska, J. Xavier; Aguilar, J.; Ahlen, S.; **Anand, Abhijeet**; *et al.*, 2025, *The Multi-Phase Circumgalactic Medium of DESI Emission-Line Galaxies at z 1.5*, ArXiv, arXiv:2511.03195 (arXiv:2511.03195) [6 citations] **[Impact Factor: N/A; Scopus Quartile: N/A]**

My role: Contributed to C IV column-density measurements and to interpreting the results in the context of previous CGM studies.

5 Wu, Xuanyi; Cai, Z.; Lan, T.-W.; Zou, S.; **Anand, Abhijeet**; *et al.*, 2025, *Tracing the Evolution of the Cool Gas in CGM and IGM Environments through Mg II Absorption from Redshift z = 0.75 to z = 1.65 Using DESI-Y1 Data*, The Astrophysical Journal, **983**, 186 (arXiv:2407.17809) [16 citations] **[Impact Factor: 5.7; Scopus Quartile: Q1]**

My role: Contributed to the discussion and interpretation of the results in the context of the CGM of emission-line galaxies.

4 Chang, Yu-Ling; Lan, Ting-Wen; Prochaska, J. Xavier; Napolitano, Lucas; **Anand, Abhijeet**; *et al.*, 2024, *Probing the Impact of Radio-mode Feedback on the Properties of the Cool Circumgalactic Medium*, The Astrophysical Journal, **974**, 191 (arXiv:2405.08314) [9 citations] **[Impact Factor: 5.7; Scopus Quartile: Q1]**

My role: Contributed to the writing and interpretation of gas profiles around luminous red galaxies in the context of previous studies.

3 Galiullin, Ilkham; Rodriguez, Antonio C.; El-Badry, Kareem; Szkody, Paula; **Anand, Abhijeet**; *et al.*, 2024, *Searching for new cataclysmic variables in the Chandra Source Catalog*, Astronomy and Astrophysics, **690**, A374 (arXiv:2408.00078) [9 citations] **[Impact Factor: N/A; Scopus Quartile: N/A]**

My role: Contributed to cross-matching Chandra sources with SDSS and DESI EDR data, leading to the identification of new cataclysmic variables.

2 Napolitano, Lucas; Pandey, Agnesh; Myers, Adam D.; Lan, Ting-Wen; **Anand, Abhijeet**; *et al.*, 2023, *Detecting and Characterizing Mg II Absorption in DESI Survey Validation Quasar Spectra*, The Astronomical Journal, **166**, 99 (arXiv:2305.20016) [34 citations] **[Impact Factor: 5.8; Scopus Quartile: Q1]**

My role: Contributed to algorithm development and to the physical interpretation of Mg II absorbers detected in early DESI quasar data.

1 Ayromlou, Mohammadreza; Kauffmann, Guinevere; **Anand, Abhijeet**; & White, Simon D. M., 2023, *The physical origin of galactic conformity: from theory to observation*, Monthly Notices of the Royal Astronomical Society, **519**, 1913 (arXiv:2207.02218) [33 citations] **[Impact Factor: 5.2; Scopus Quartile: Q1]**

My role: Contributed to comparing galactic conformity in simulations with observations from the Legacy Imaging Surveys and SDSS.

Collaboration papers (citations: 7791)

39 Chang, Yu-Ling; Lan, Ting-Wen; Prochaska, J. Xavier; Siudek, Malgorzata; Aguilar, J.; *et al.* (incl. **A. Anand**), 2026, *Tracing the Cosmic Evolution of the Cool Circumgalactic Medium of Luminous Red Galaxies with DESI Year 1 Data*, The Astrophysical Journal, **1009**, 121 (arXiv:2512.03845) [4 citations] [**Impact Factor: 5.7; Scopus Quartile: Q1**]

38 Lange, J. U.; Wells, A.; Hearin, A.; Beltz-Mohrmann, G.; Leauthaud, A.; *et al.* (incl. **A. Anand**), 2026, *Cosmological Constraints from Full-scale Clustering and Galaxy-Galaxy Lensing with DESI DR1*, The Astrophysical Journal, **1010**, 170 (arXiv:2512.15962) [**Impact Factor: 5.7; Scopus Quartile: Q1**]

37 Uzsoy, Ana Sofía M.; Dey, Arjun; Raichoor, Anand; Finkbeiner, Douglas P.; Ramakrishnan, Vandana; *et al.* (incl. **A. Anand**), 2026, *Effect of Local Environment on Lyα Line Profile in DESI/ODIN LAEs*, The Astrophysical Journal, **1010**, 76 (arXiv:2511.17498) [**Impact Factor: 5.7; Scopus Quartile: Q1**]

36 Blanco, Diana; Leauthaud, Alexie; Lange, Johannes Ulf; Wright, Angus H.; Hildebrandt, Hendrik; *et al.* (incl. **A. Anand**), 2026, *The Power of DESI for Photometric Redshift Calibration: A Case Study with KiDS-1000*, The Open Journal of Astrophysics, **9**, 63965 (arXiv:2512.15964) [3 citations] [**Impact Factor: 2.4; Scopus Quartile: Q1**]

35 McArthur, E.; Millon, M.; Powell, M.; Wechsler, R. H.; Pan, Z.; *et al.* (incl. **A. Anand**), 2026, *Quasars Acting as Strong Lenses Found in DESI DR1*, The Astrophysical Journal, **1006**, 147 (arXiv:2511.02009) [1 citations] [**Impact Factor: 5.7; Scopus Quartile: Q1**]

34 Saydjari, Andrew K.; Draine, Bruce T.; Brandt, Timothy D.; Schlafly, Edward F.; Dey, Arjun; *et al.* (incl. **A. Anand**), 2026, *Spatially-Resolved Spectra of Diffuse Galactic Light using 10.8 M DESI Sky Fibers*, ArXiv, arXiv:2608.28740 (arXiv:2608.28740) [**Impact Factor: N/A; Scopus Quartile: N/A**]

33 Crespi, Andrea; Percival, Will J.; Krolewski, Alex; Bonici, Marco; Zhang, Hanyu; *et al.* (incl. **A. Anand**), 2026, *Baryon fraction from the BAO amplitude: A consistent approach to parametrizing perturbation growth*, Physical Review D, **114**, 043545 (arXiv:2511.23459) [1 citations] [**Impact Factor: 5.3; Scopus Quartile: Q1**]

32 Yin, Gaocheng; Jiang, Linhua; Pan, Zhiwei; Martini, Paul; Guo, Wei-Jian; *et al.* (incl. **A. Anand**), 2026, *Luminosity function of quasars at* $1.0 < z < 3.5$ *from SDSS and DESI*, ArXiv, arXiv:2608.06000 (arXiv:2608.06000) [**Impact Factor: N/A; Scopus Quartile: N/A**]

31 DESI Collaboration; Adame, A. G.; Aguilar, J.; Ahlen, S.; Alves, O.; *et al.* (incl. **A. Anand**), 2026, *DESI DR2 Results IV: Alcock-Paczyński Measurements from the Lyman Alpha Forest and Cosmological Constraints*, ArXiv, arXiv:2607.27410 (arXiv:2607.27410) [9 citations] [**Impact Factor: N/A; Scopus Quartile: N/A**]

30 Dey, Biprateep; Newman, Jeffrey A.; Zhang, Tianqing; Aguilar, J.; Ahlen, S.; *et al.* (incl. **A. Anand**), 2026, *Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration*, The Astronomical Journal, **172**, 65 (arXiv:2604.06143) [2 citations] [**Impact Factor: 5.8; Scopus Quartile: Q1**]

29 Moore, Samuel G.; Cole, Shaun; Wilson, Michael; Norberg, Peder; Moustakas, John; *et al.* (incl. **A. Anand**), 2026, *DESI DR2 Galaxy luminosity functions*, Monthly Notices of the Royal

Astronomical Society, **549**, stag866 (arXiv:2511.01803) [10 citations] [**Impact Factor: 5.2; Scopus Quartile: Q1**]

28 Negi, Vibhore; Joshi, Ravi; Vishnupriya, B.; Wu, Xue-Bing; Yesuf, Hassen M.; *et al.* (incl. **A. Anand**), 2026, *DESI as sparse Integral Field Spectrograph I: Spatially resolved chemical enrichment in star-forming galaxies at $z \leq 0.1$*, ArXiv, arXiv:2606.12541 (arXiv:2606.12541) [**Impact Factor: N/A; Scopus Quartile: N/A**]

27 DESI Collaboration; Abdul Karim, M.; Adame, A. G.; Aguado, D.; Aguilar, J.; *et al.* (incl. **A. Anand**), 2026, *Data Release 1 of the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **171**, 285 (arXiv:2503.14745) [705 citations] [**Impact Factor: 5.8; Scopus Quartile: Q1**]

26 Turner, Wynne; Cuceu, Andrei; Martini, Paul; Aguilar, J.; Ahlen, S.; *et al.* (incl. **A. Anand**), 2026, *Probing the limits of cosmological information from the Lyman-α forest 2-point correlation functions*, Journal of Cosmology and Astroparticle Physics, **2026**, 010 (arXiv:2509.14322) [4 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

25 Zaborowski, E. A.; Taylor, P.; Honscheid, K.; Cuceu, A.; de Mattia, A.; *et al.* (incl. **A. Anand**), 2026, *H ₀ without the sound horizon (or supernovae): A 2% measurement in DESI DR1*, Journal of Cosmology and Astroparticle Physics, **2026**, 004 (arXiv:2510.19149) [10 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

24 Karaçaylı, Naim Göksel; Ravoux, Corentin; Martini, Paul; Goff, Jean-Marc Le; Armengaud, Eric; *et al.* (incl. **A. Anand**), 2026, *DESI DR1 Lyα 1D power spectrum: Validation of estimators*, Journal of Cosmology and Astroparticle Physics, **2026**, 048 (arXiv:2509.13593) [6 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

23 Ebina, H.; White, M.; Raichoor, A.; Dey, Arjun; Schlegel, D.; *et al.* (incl. **A. Anand**), 2026, *Clustering analysis of medium-band selected high-redshift galaxies*, Journal of Cosmology and Astroparticle Physics, **2026**, 019 (arXiv:2509.26467) [7 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

22 Hsu, Yuan-Ming; Huang, Xiaosheng; Storfer, Christopher J.; Inchausti, Jose Carlos; Schlegel, David; *et al.* (incl. **A. Anand**), 2026, *A New Way to Discover Strong Gravitational Lenses: Pairwise Spectroscopic Search from DESI DR1*, The Astrophysical Journal Supplement Series, **282**, 41 (arXiv:2509.16033) [5 citations] [**Impact Factor: 7.7; Scopus Quartile: Q1**]

21 Han, Jiwon Jesse; Chiti, Anirudh; Chen, Kai-Feng; Bechtol, Keith; Bellini, Andrea; *et al.* (incl. **A. Anand**), 2026, *A Path to an All-Sky Survey with Roman*, ArXiv, arXiv:2602.21280 (arXiv:2602.21280) [3 citations] [**Impact Factor: N/A; Scopus Quartile: N/A**]

20 Kamalinejad, Farshad; Slepian, Zachary; Krolewski, Alex; Greco, Alessandro; Ortolá Leonard, William; *et al.* (incl. **A. Anand**), 2026, *First Detection of the Baryon Acoustic Oscillation (BAO) Feature in the 3-Point Correlation Function of DESI DR1 Luminous Red Galaxies*, ArXiv, arXiv:2602.16134 (arXiv:2602.16134) [4 citations] [**Impact Factor: N/A; Scopus Quartile: N/A**]

19 Herrera-Alcantar, Hiram K.; Armengaud, Eric; Yèche, Christophe; Gordon, Calum; Casas, Laura; *et al.* (incl. **A. Anand**), 2025, *The Lyman-α forest from LBGs: First 3D correlation measurement with DESI and prospects for cosmology*, Journal of Cosmology and Astroparticle Physics, **2025**, 053 (arXiv:2507.21852) [8 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

18 Abdul Karim, M.; Aguilar, J.; Ahlen, S.; Alam, S.; Allen, L.; *et al.* (incl. **A. Anand**), 2025, *DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints*,

Physical Review D, **112**, 083515 (arXiv:2503.14738) [1712 citations] [**Impact Factor: 5.3; Scopus Quartile: Q1**]

17 Abdul Karim, M.; Aguilar, J.; Ahlen, S.; Allende Prieto, C.; Alves, O.; *et al.* (incl. **A. Anand**), 2025, *DESI DR2 results. I. Baryon acoustic oscillations from the Lyman alpha forest*, Physical Review D, **112**, 083514 (arXiv:2503.14739) [333 citations] [**Impact Factor: 5.3; Scopus Quartile: Q1**]

16 Brodzeller, A.; Wolfson, M.; Santos, D. M.; Ho, M.; Tan, T.; *et al.* (incl. **A. Anand**), 2025, *Construction of the damped Lyα absorber catalog for DESI DR2 Lyα BAO*, Physical Review D, **112**, 083510 (arXiv:2503.14740) [30 citations] [**Impact Factor: 5.3; Scopus Quartile: Q1**]

15 Rashkovetskyi, Michael; Eisenstein, Daniel J.; Aguilar, Jessica Nicole; Ahlen, Steven; **Anand, Abhijeet**; *et al.*, 2025, *Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect*, The Open Journal of Astrophysics, **8**, 46033 (arXiv:2508.20904) [**Impact Factor: 2.4; Scopus Quartile: Q1**]

14 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 V: Full-Shape galaxy clustering from galaxies and quasars*, Journal of Cosmology and Astroparticle Physics, **2025**, 008 (arXiv:2411.12021) [211 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

13 Treiber, Helena; Amon, Alexandra; Wechsler, Risa H.; Manwadkar, Viraj; Myles, Justin; *et al.* (incl. **A. Anand**), 2025, *Dwarf galaxy halo masses from spectroscopic and photometric lensing in DESI and DES*, ArXiv, arXiv:2509.20434 (arXiv:2509.20434) [4 citations] [**Impact Factor: N/A; Scopus Quartile: N/A**]

12 Cuceu, Andrei; Herrera-Alcantar, Hiram K.; Gordon, Calum; Ramírez-Pérez, César; Armengaud, E.; *et al.* (incl. **A. Anand**), 2025, *DESI DR1 Lyα forest: 3D full-shape analysis and cosmological constraints*, ArXiv, arXiv:2509.15308 (arXiv:2509.15308) [29 citations] [**Impact Factor: N/A; Scopus Quartile: N/A**]

11 Medina, Gustavo E.; Li, Ting S.; Eadie, Gwendolyn M.; Riley, Alexander H.; Valluri, Monica; *et al.* (incl. **A. Anand**), 2025, *The mass of the Milky Way from outer halo stars measured by DESI DR1*, ArXiv, arXiv:2508.19351 (arXiv:2508.19351) [9 citations] [**Impact Factor: N/A; Scopus Quartile: N/A**]

10 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 VII: cosmological constraints from the full-shape modeling of clustering measurements*, Journal of Cosmology and Astroparticle Physics, **2025**, 028 (arXiv:2411.12022) [410 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

9 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 II: sample definitions, characteristics, and two-point clustering statistics*, Journal of Cosmology and Astroparticle Physics, **2025**, 017 (arXiv:2411.12020) [156 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

8 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 III: baryon acoustic oscillations from galaxies and quasars*, Journal of Cosmology and Astroparticle Physics, **2025**, 012 (arXiv:2404.03000) [592 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

7 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations*, Journal of Cosmology and Astroparticle Physics, **2025**, 021 (arXiv:2404.03002) [2020

citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

6 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 IV: Baryon Acoustic Oscillations from the Lyman alpha forest*, Journal of Cosmology and Astroparticle Physics, **2025**, 124 (arXiv:2404.03001) [417 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

5 Ross, A. J.; Aguilar, J.; Ahlen, S.; Alam, S.; **Anand, Abhijeet**; *et al.*, 2025, *The construction of large-scale structure catalogs for the Dark Energy Spectroscopic Instrument*, Journal of Cosmology and Astroparticle Physics, **2025**, 125 (arXiv:2405.16593) [79 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

4 Scholte, Dirk; Saintonge, Amélie; Moustakas, John; Catinella, Barbara; Zou, Hu; *et al.* (incl. **A. Anand**), 2024, *The atomic gas sequence and mass-metallicity relation from dwarfs to massive galaxies*, Monthly Notices of the Royal Astronomical Society, **535**, 2341 (arXiv:2408.03996) [37 citations] [**Impact Factor: 5.2; Scopus Quartile: Q1**]

3 DESI Collaboration; Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; *et al.* (incl. **A. Anand**), 2024, *The Early Data Release of the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **168**, 58 (arXiv:2306.06308) [620 citations] [**Impact Factor: 5.8; Scopus Quartile: Q1**]

2 DESI Collaboration; Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; *et al.* (incl. **A. Anand**), 2024, *Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **167**, 62 (arXiv:2306.06307) [339 citations] [**Impact Factor: 5.8; Scopus Quartile: Q1**]

1 Han, Jiwon Jesse; Dey, Arjun; Price-Whelan, Adrian M.; Najita, Joan; Schlafly, Edward F.; *et al.* (incl. **A. Anand**), 2023, *NANCY: Next-generation All-sky Near-infrared Community surveY*, ArXiv, arXiv:2306.11784 (arXiv:2306.11784) [11 citations] [**Impact Factor: N/A; Scopus Quartile: N/A**]